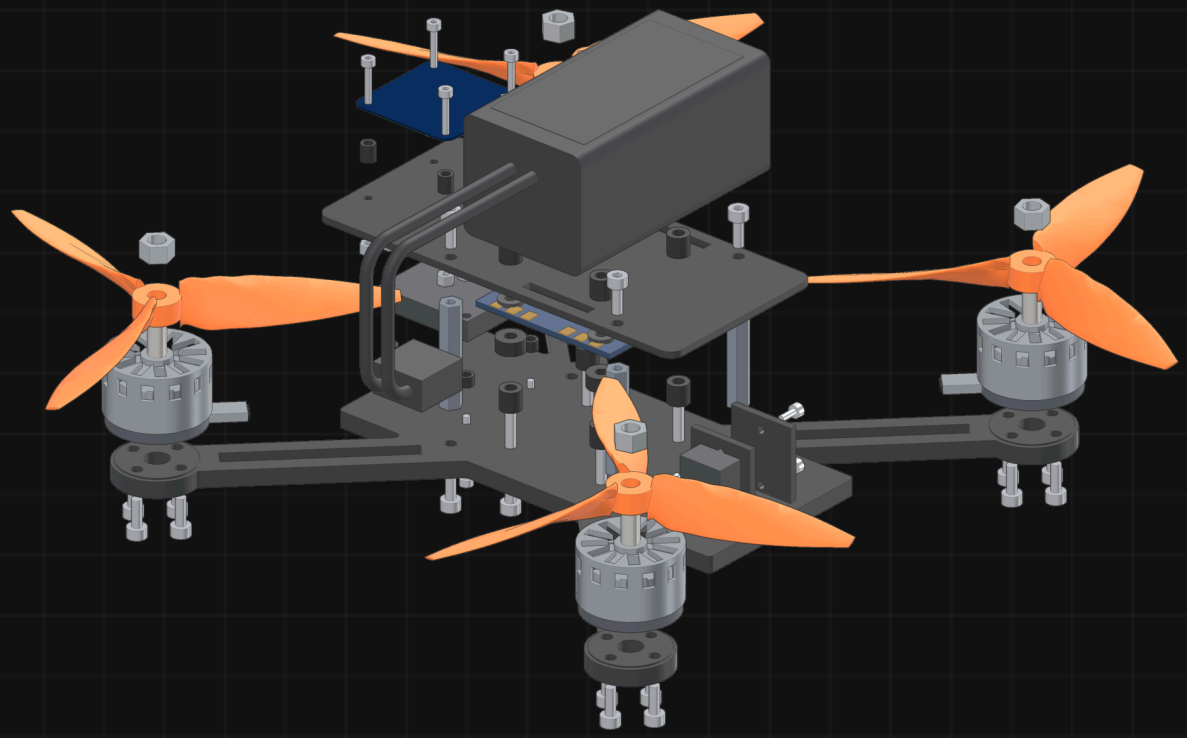
Rev A | 2026-10-24 | UHD-QUAD5-AG

ASSEMBLY GUIDE

5-inch 6S Quadcopter

Reference 5-inch 6S freestyle quadcopter: DolphinRC F405 V3 + AM32 60A stack, four MEPS NEON 2207 V2 1950KV motors with HQProp Ethix S5 props, CNHL 1100 mAh 6S pack, RadioMaster RP1 V2 ELRS receiver, DJI O4 Air Unit, Matek M9N-5883 GNSS/compass, custom frame.



Exploded along each joint's axis in fastener-stack order (14 mates, 84 hardware parts, battery by placement hint).

STEPS	MATES	HARDWARE	PARTS
12	14	84	118
mechanical and wiring	from assemble()	screws, spacers, standoffs, nuts	43 BOM lines

HOW THIS GUIDE IS MADE

Steps, parts and hardware come from the UHD model: mates and fastener stacks from `assemble()`, wiring from `wiringChecklist`. The coverage table at the end lists where every mate, fastener and wire appears.

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

01 Before you begin

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TOOLS

- **1.5 mm hex key** for M2 × 18 mm SHCS, M2 × 5 mm SHCS, M2 × 12 mm SHCS (ISO 4762 socket size)
- **2.5 mm hex key** for M3 × 30 mm SHCS, M3 × 8 mm SHCS, M3 × 8 mm motor screw
- **4 mm nut driver** for M2 nyloc nut, DIN 985, A2 stainless
- **5.5 mm nut driver** for M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless
- **8 mm nut driver** for M5 flanged lock nut (prop nut, included with the motor)
- **5.5 mm spanner** to hold M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF
- Soldering iron, solder, heat-shrink

Derived from the fasteners in the model: hex keys from each screw's stated socket or the ISO 4762 socket size for its thread, drivers and spanners from each nut's and standoff's across-flats.

READING THE FIGURES

Already assembled
Fitted in this step
Hardware for this step
Assembly axis

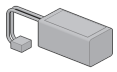
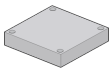
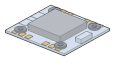
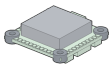
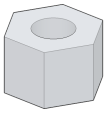
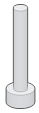
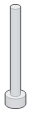
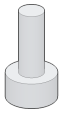
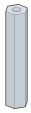
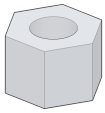
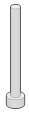
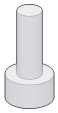
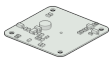
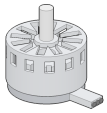
Parts are drawn exploded along each joint's axis in stacking order, from the model's fastener stacks and mate gaps.

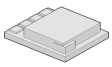
⚠ **WARNING** Remove the propellers for any work with the battery connected. Fit them last (step 12).

ⓘ **NOTE** Each step's torque line gives the tightening torque of the part turned, from the harness's fastener stack. 9 of them are **assumptions** (marked A): no frame, motor or DJI source states a torque, so they are derated from supplier tables for the screw grade into steel. Threadlocker is only advised where a harness says so (steel into aluminium).

02 Kit contents

DOCS/QUADCOPTER-5IN-BOM.JSON

no CAD **×1** Cable ties, 2.5 mm × 100 mm quad-cable-ties	no CAD **×1** Clear heat-shrink sleeve for the receiver, about 15 mm flat width quad-receiver-heat-shrink	 **×1** CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) cnhl-black-series-1100mah-6s-100c	 **×1** DJI O4 Air Unit dji-o4-air-unit	 **×1** DolphinRC AM32 60A 4-in-1 ESC dolphinrc-am32-60a-4in1-esc
 **×1** DolphinRC F405 V3 Flight Controller dolphinrc-f405-v3-flight-controller	no CAD **×1** Double-sided foam mounting tape, 1 mm quad-foam-mount-tape	 **×8** ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 ettinger-005-83-060-m3-nylon-spacer-6mm	no CAD **×1** Flux-cored solder wire, 0.8 mm quad-solder-wire	no CAD **×1** HGLRC Kevlar battery strap 20x250mm hglrc-kevlar-battery-strap-20x250mm
 **×4** HQProp Ethix S5 5x4x3 Propeller hqprop-ethix-s5	 **×8** M2 nyloc nut, DIN 985, A2 stainless iso-10511-m2-nyloc-nut	 **×4** M2 nylon spacer, Ø2.7/4.0 x 3 mm m2-nylon-spacer-3mm	 **×4** M2 nylon spacer, Ø2.7/4.0 x 5 mm m2-nylon-spacer-5mm	 **×4** M2 x 12 mm SHCS iso-4762-m2x12
 **×4** M2 x 18 mm SHCS iso-4762-m2x18	 **×4** M2 x 5 mm SHCS iso-4762-m2x5	 **×4** M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF m3-aluminium-standoff-30mm	 **×4** M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless iso-10511-m3-nyloc-nut	 **×4** M3 x 30 mm SHCS iso-4762-m3x30-socket-head-cap-screw
 **×8** M3 x 8 mm SHCS iso-4762-m3x8	 **×1** Matek M9N-5883 GNSS & Compass matek-m9n-5883	 **×4** MEPS NEON 2207 V2 1950KV Brushless Motor meps-neon-2207-v2-1950kv	no CAD **×1** No-clean flux (pen or paste) quad-flux	 **×1** Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor panasonic-eeu-fr1v471



×1

RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver

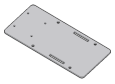
radiomaster-rp1-v2-elrs-2g4

no CAD

×1

Threadlocker, medium strength (blue)

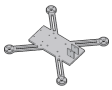
quad-threadlocker



×1

5-inch frame top plate

quadcopter-5in-top-plate



×1

5-inch quadcopter frame

quadcopter-5in-frame



×1

Bulk capacitor leads

panasonic-eeu-fr1v471-leads



×1

DJI O4 3-in-1 cable

dji-o4-3in1-cable



×1

FC/ESC SH1.0 8-pin cable

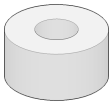
dolphinrc-sh8-fc-esc-cable



×1

GNSS lead (GH6P, FC end soldered)

matek-gh6p-gnss-lead



×4

M3 rubber grommet, ESC

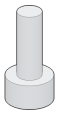
dolphinrc-m3-grommet-esc



×4

M3 rubber grommet, FC

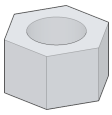
dolphinrc-m3-grommet-fc



×16

M3 × 8 mm motor screw

motor-screw-m3x8



×4

M5 flanged lock nut (prop nut, included with the motor)

meps-m5-flanged-lock-nut



×1

Motor 1 phase leads (Rear-right)

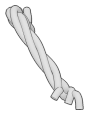
meps-phase-leads-m1



×1

Motor 2 phase leads (Front-right)

meps-phase-leads-m2



×1

Motor 3 phase leads (Rear-left)

meps-phase-leads-m3



×1

Motor 4 phase leads (Front-left)

meps-phase-leads-m4



×1

Receiver wires (RP1 V2 CRSF wire)

radiomaster-rp1-crsf-wire



×1

XT60 battery lead (12 AWG)

dolphinrc-xt60-battery-lead

Quantities from the generated BOM; drawings are the parts' CAD artifacts, not to scale.

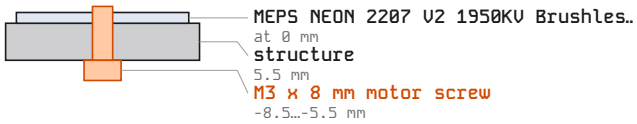
03 Hardware

FASTENER STACKS

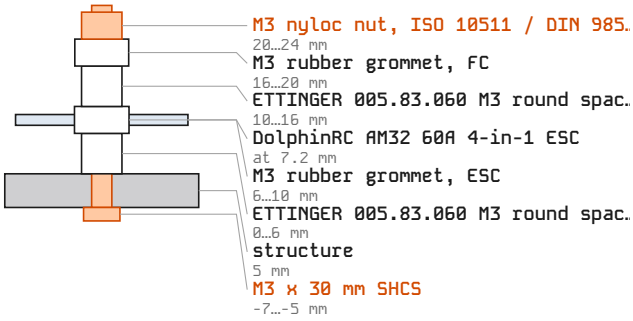
Every screw, spacer, standoff and nut is placed by a harness's `fastenerStack`. The sections below are drawn from those positions and the mate gaps: one hole of each joint, to one scale, millimetres from the structure face.

HARNESS	PARTS PLACED	TORQUE	USED IN
Motor hardware (M3) ×4 joint arm_fr_mount, arm_rr_mount, arm_rl_mount, arm_fl_mount	16 × M3 × 8 mm motor screw	0.6 N·m[A] M3 × 8 mm motor screw	Step 1
Stack hardware (M3) joint stack_mount	4 × M3 × 30 mm SHCS 8 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 4 × M3 rubber grommet, ESC 4 × M3 rubber grommet, FC 4 × M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	0.3 N·m[A] M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	Step 3
Frame hardware (M3 standoffs) joint top_plate_mount	8 × M3 × 8 mm SHCS 4 × M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	0.6 N·m[A] M3 × 8 mm SHCS 0.6 N·m[A] M3 × 8 mm SHCS	Step 9
Camera hardware (M2) joint camera_mount	4 × M2 × 5 mm SHCS	0.1 N·m[A] M2 × 5 mm SHCS 0.1 N·m[A] M2 × 5 mm SHCS	Step 5
Air unit hardware (M2) joint vtx_mount	4 × M2 × 18 mm SHCS 4 × M2 nylon spacer, Ø2.7/4.0 × 3 mm 4 × M2 nyloc nut, DIN 985, A2 stainless	0.1 N·m[A] M2 nyloc nut, DIN 985, A2 stainless	Step 5
GPS hardware (M2) joint gps_mount	4 × M2 nyloc nut, DIN 985, A2 stainless 4 × M2 nylon spacer, Ø2.7/4.0 × 5 mm 4 × M2 × 12 mm SHCS	0.1 N·m[A] M2 × 12 mm SHCS	Step 10
Prop nut (M5) ×4 joint arm_fr/prop_on_shaft, arm_rr/prop_on_shaft, arm_rl/prop_on_shaft, arm_fl/prop_on_shaft	4 × M5 flanged lock nut (prop nut, included with the motor)	1 N·m[A] M5 flanged lock nut (prop nut, included with the motor)	Step 12

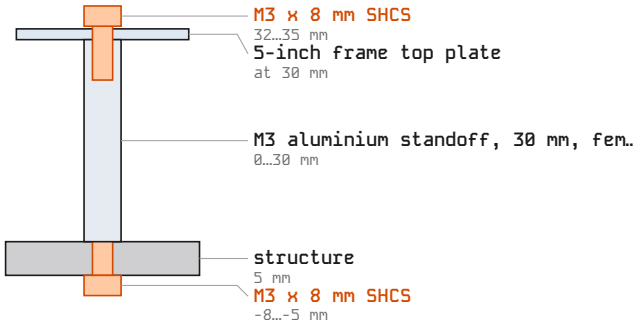
Motor hardware (M3) quadcopter-5in-motor-hardware



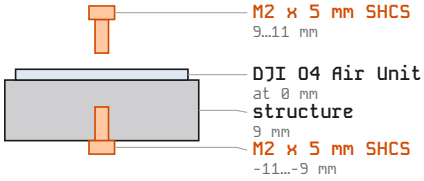
Stack hardware (M3) quadcopter-5in-stack-hardware



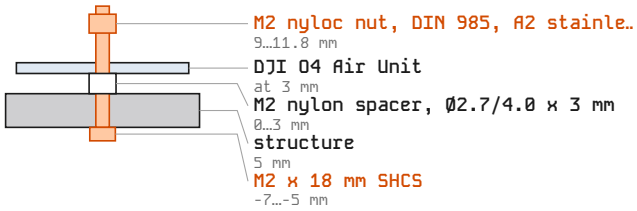
Frame hardware (M3 standoffs) quadcopter-5in-frame-hardware



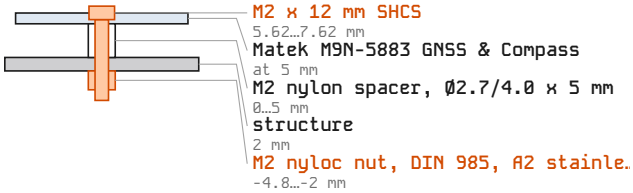
Camera hardware (M2) quadcopter-5in-camera-hardware



Air unit hardware (M2) quadcopter-5in-vtx-hardware



GPS hardware (M2) quadcopter-5in-gps-hardware



Prop nut (M5) quadcopter-5in-prop-hardware



1 Mount the motors

mates: arm_fr_mount, arm_rr_mount, arm_rl_mount, arm_fl_mount



A ×4
MEPS NEON 2207 V2 1950KV Brushless Motor



B ×16
M3 × 8 mm motor screw

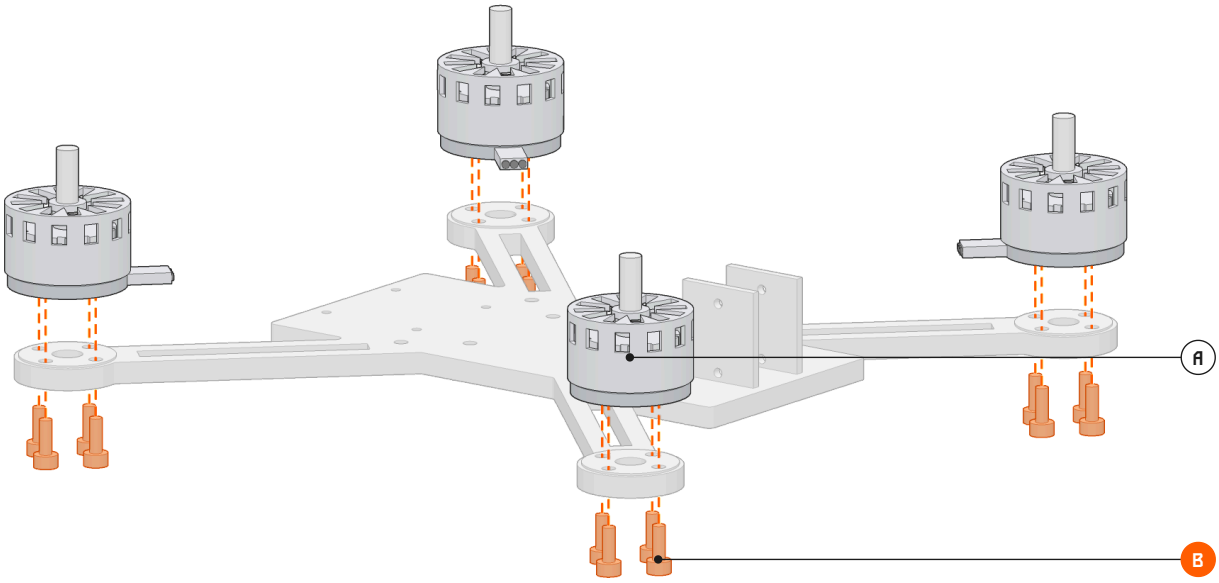


Figure 1 Mount the motors: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. From each motor's box take the four screws that reach 2–3 mm into the motor base through the 5.5 mm arm and pad (8 mm; MEPS ships two lengths and states neither): a screw that goes deeper can touch the windings.
2. Insert 4 × M3 × 8 mm motor screw from below, up through the motor mount.
3. Seat a MEPS NEON 2207 V2 1950KV Brushless Motor onto each motor mount (turned 180° / 270° / 90° by corner, per the mates). Repeat on all 4.

Torque 0.6 N·m[A] M3 × 8 mm motor screw **Tools** 2.5 mm hex key **Step** 1 of 12

TIP Point each motor's leads back along its arm toward the stack; the rotation on the pad is set by the mate.

TIP Threadlocker: medium strength (blue) on the motor screws. Steel into aluminium with no locking element, so vibration can back the screws out. Advice is from X-Team's motor-screw guide, a secondary source; no maker of these parts states it. — quadcopter-5in-motor-hardware

2 Connect the FC and ESC

links: stack/fc_esc_plug_fc, stack/fc_esc_plug_esc



Ⓐ ×1

FC/ESC SH1.0 8-pin cable



Figure 2 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Connect FC ESC socket, ESC FC socket as in the table (16 connections).

LINK	FROM	TO	TERMINATION	DRC
FC ESC socket	Flight controller **VBAT · GND · CUR · P4 · M1 · M2 · M3 · M4**	SH1.0 8-pin cable **P1 · P2 · P3 · P4 · P5 · P6 · P7 · P8**	jst_sh_8 / jst_sh_8	CONFIGURED
ESC FC socket	SH1.0 8-pin cable **P1 · P2 · P3 · P4 · P5 · P6 · P7 · P8**	4-in-1 ESC **V · G · CUR · P4 · M1 · M2 · M3 · M4**	jst_sh_8 / jst_sh_8	CONFIGURED

Step 2 of 12

ⓘ **NOTE** Included with the DolphinRC F405 V3 stack (manual: 'SH1.0mm 8Pin Cable 50mm'). Straight-through, so pin 1 maps to pin 1. — dolphinrc-sh8-fc-esc-cable

3 Mount the stack on the frame

mates: stack_mount, fc_mount



(A) ×1
DolphinRC AM32 60A 4-in-1 ESC



(B) ×1
DolphinRC F405 V3 Flight Controller



C ×4
M3 × 30 mm SHCS



D ×8
ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25



E ×4
M3 rubber grommet, ESC



F ×4
M3 rubber grommet, FC



G ×4
M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless

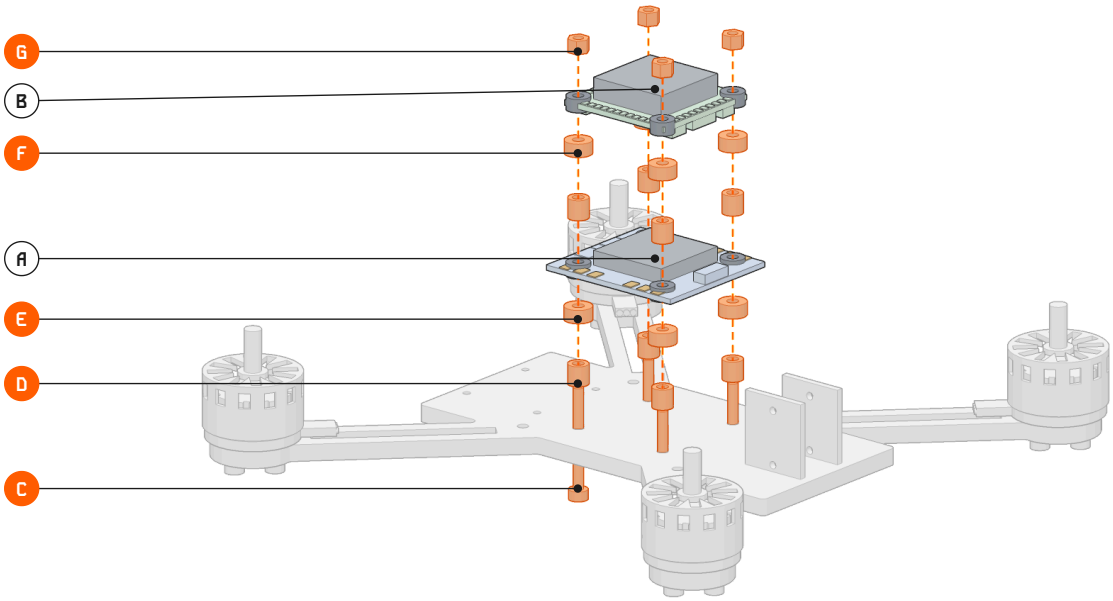


Figure 3 Mount the stack on the frame: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Keep the FC/ESC cable plugged in, as the stack was tested: lower the ESC and then the FC over the screws together.
2. Insert 4 × M3 × 30 mm SHCS from below, up through the stack mount.
3. Slide 4 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 over the screws.
4. Slide 4 × M3 rubber grommet, ESC over the screws.
5. Lower the DolphinRC AM32 60A 4-in-1 ESC onto the stack mount (7.2 mm gap).
6. Slide 4 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25 over the screws.
7. Slide 4 × M3 rubber grommet, FC over the screws.
8. Lower the DolphinRC F405 V3 Flight Controller onto the stack mount (17.2 mm gap).
9. Finish with 4 × M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless; tighten until snug.

Torque 0.3 N·m[A] M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless **Tools** 2.5 mm hex key, 5.5 mm nut driver **Step** 3 of 12

TIP Battery pads to the rear. Seen from above, the ESC's motor pads are then at their motors' corners: M1 rear-right, M2 front-right, M3 rear-left, M4 front-left (docs/decisions.md D-Q02).

NOTE Per post, bottom to top: screw head under the 5 mm bottom plate, 6 mm spacer, ESC in its grommet (4 mm), 6 mm spacer, FC in its grommet (4 mm), nyloc nut; 1 mm of thread protrudes. Boards sit in the M3 grommets included with the DolphinRC stack (dolphinrc-m3-grommet-esc, -fc; four each, one spare each in the box), which the stack places. Stack-up worked out in the fastener parts' usage notes (assumes a 5 mm plate and 4 mm grommet clamp). — quadcopter-5in-stack-hardware

4 Solder the motor leads

links: m2_leads, m2_pads, m1_leads, m1_pads, m3_leads, m3_pads, m4_leads, m4_pads



Ⓐ ×1
Motor 2 phase leads (Front-right)



Ⓑ ×1
Motor 1 phase leads (Rear-right)



Ⓒ ×1
Motor 3 phase leads (Rear-left)



Ⓓ ×1
Motor 4 phase leads (Front-left)

Figure 4 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Connect Motor 2 leads, Motor 2 pads, Motor 1 leads, Motor 1 pads, Motor 3 leads, Motor 3 pads, Motor 4 leads, Motor 4 pads as in the table (24 connections).

LINK	FROM	TO	TERMINATION	DRC
Motor 2 leads	arm_fr **A · B · C**	Motor 2 phase leads **P1 · P2 · P3**	integral	CONFIGURED
Motor 2 pads	Motor 2 phase leads **P1 · P2 · P3**	stack **M2 P1 · M2 P2 · M2 P3**	solder_leads	CONFIGURED
Motor 1 leads	arm_rr **A · B · C**	Motor 1 phase leads **P1 · P2 · P3**	integral	CONFIGURED
Motor 1 pads	Motor 1 phase leads **P1 · P2 · P3**	stack **M1 P1 · M1 P2 · M1 P3**	solder_leads	CONFIGURED
Motor 3 leads	arm_rl **A · B · C**	Motor 3 phase leads **P1 · P2 · P3**	integral	CONFIGURED
Motor 3 pads	Motor 3 phase leads **P1 · P2 · P3**	stack **M3 P1 · M3 P2 · M3 P3**	solder_leads	CONFIGURED
Motor 4 leads	arm_fl **A · B · C**	Motor 4 phase leads **P1 · P2 · P3**	integral	CONFIGURED
Motor 4 pads	Motor 4 phase leads **P1 · P2 · P3**	stack **M4 P1 · M4 P2 · M4 P3**	solder_leads	CONFIGURED

Step 4 of 12

TIP Any two leads of a motor may be swapped to reverse it. Set each motor to the spin given in the propeller step, in the ESC configurator rather than by re-soldering.

NOTE The motor's own leads: 150 ± 2 mm, 10 mm heat shrink at the motor, 3 mm pre-tinned tips; three loose white silicone leads (MEPS page and photos). Cut to length, twisted to keep them tidy and reduce interference (Oscar Liang), soldered to the pads; swap any two to reverse the motor. — meps-phase-leads-m2

NOTE The motor's own leads: 150 ± 2 mm, 10 mm heat shrink at the motor, 3 mm pre-tinned tips; three loose white silicone leads (MEPS page and photos). Cut to length, twisted to keep them tidy and reduce interference (Oscar Liang), soldered to the pads; swap any two to reverse the motor. — meps-phase-leads-m1

NOTE The motor's own leads: 150 ± 2 mm, 10 mm heat shrink at the motor, 3 mm pre-tinned tips; three loose white silicone leads (MEPS page and photos). Cut to length, twisted to keep them tidy and reduce interference (Oscar Liang), soldered to the pads; swap any two to reverse the motor. — meps-phase-leads-m3

NOTE The motor's own leads: 150 ± 2 mm, 10 mm heat shrink at the motor, 3 mm pre-tinned tips; three loose white silicone leads (MEPS page and photos). Cut to length, twisted to keep them tidy and reduce interference (Oscar Liang), soldered to the pads; swap any two to reverse the motor. — meps-phase-leads-m4

5 Install the air unit and camera

mates: camera_mount, vtx_mount

A ×1 DJI O4 Air Unit	B ×4 M2 × 5 mm SHCS	C ×4 M2 × 18 mm SHCS	D ×4 M2 nylon spacer, Ø2.7/4.0 × 3 mm

E ×4
M2 nyloc nut, DIN 985, A2 stainless

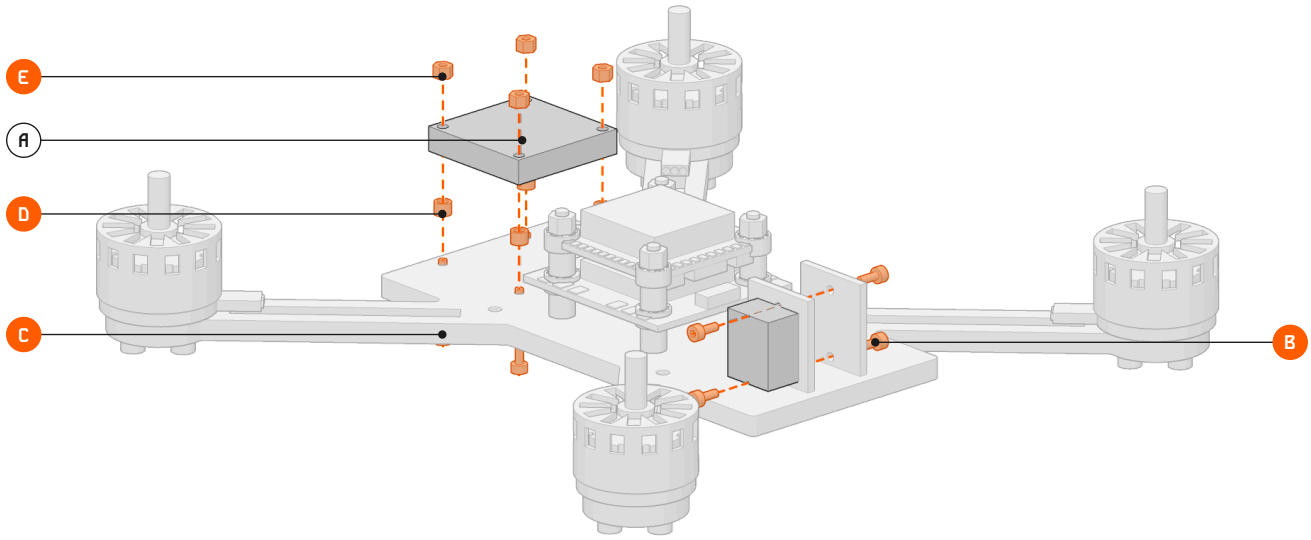


Figure 5 Install the air unit and camera: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Insert 2 × M2 × 5 mm SHCS through the camera side plates from one side.
2. Seat the DJI O4 Air Unit onto the camera side plates.
3. Insert 2 × M2 × 5 mm SHCS from the opposite side.
4. Insert 4 × M2 × 18 mm SHCS from below, up through the transmission module mount.
5. Slide 4 × M2 nylon spacer, Ø2.7/4.0 × 3 mm over the screws.
6. Lower the DJI O4 Air Unit onto the transmission module mount (3 mm gap).
7. Finish with 4 × M2 nyloc nut, DIN 985, A2 stainless; tighten until snug.
8. Fit the air unit's antenna and lead it out of the rear of the frame, at least 5 cm from the transmission and camera modules (DJI O4 manual); tie its lead to the frame with a cable tie where it leaves the plates.

Torque 0.1 N·m[A] M2 × 5 mm SHCS · Camera hardware (M2); 0.1 N·m[A] M2 nyloc nut, DIN 985, A2 stainless · Air unit hardware (M2)
Tools 1.5 mm hex key, 4 mm nut driver **Step** 5 of 12

NOTE DJI O4 manual p.6: 'The screws included with the camera can only be used for frames thicker than 2 mm', and 'DO NOT overtighten the screws to avoid stripping.' These side plates are 2 mm, so bought M2 × 5 screws are used and the included ones are spare. No threadlocker: the camera's thread material is not stated. — quadcopter-5in-camera-hardware

⚠ **CAUTION** Needs airflow. Install inside the frame in a well-ventilated spot (near air intake/exhaust), within ~1 cm of the propellers to use downwash, and not in an enclosed area; a heat-conductive pad to the carbon frame extends run time. Ground standby from cold start about 2.5 min at 25 C (ground low-power mode on); above 25 C ambient use external cooling. — dji-o4-air-unit

6 Solder the battery lead and capacitor

links: battery_plug, battery_pads, bulk_cap_can, bulk_cap_pads



Ⓐ ×1

Panasonic EEU-FR1V471 470 µF 35 V low-ESR electrolytic capacitor



Ⓑ ×1

XT60 battery lead (12 AWG)



Ⓒ ×1

Bulk capacitor leads

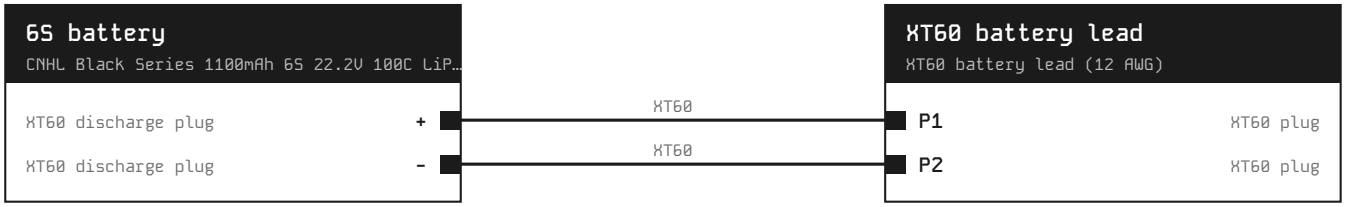


Figure 6 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Stick a 10 × 16 mm piece of foam tape on the air unit behind the ESC and lay the capacitor on it, its + lead toward the ESC's + pad.
2. Tie the XT60 lead to the frame behind the stack with a cable tie, so the plug cannot reach a propeller.
3. Connect XT60, ESC battery pads, Capacitor leads, Capacitor on the ESC battery pads as in the table (8 connections).

LINK	FROM	TO	TERMINATION	DRC
XT60	6S battery **+ · -**	XT60 battery lead **P1 · P2**	xt60 / xt60	CONFIGURED
ESC battery pads	XT60 battery lead **P1 · P2**	stack **+ · -**	solder_leads	CONFIGURED
Capacitor leads	bulk_cap **+ · -**	Bulk capacitor leads **P1 · P2**	integral	CONFIGURED
Capacitor on the ESC battery pads	Bulk capacitor leads **P1 · P2**	stack **+ · -**	solder_leads	CONFIGURED

Step 6 of 12

⚠ **CAUTION** Observe polarity on the capacitor and the XT60 lead before the first power-up.

ⓘ **NOTE** Included with the stack (product page: '12AWG 12cm red/black lead and XT60 male plug'). Cut to length before soldering to the ESC pads. Solder the bundled 470 uF capacitor across the same pads. — dolphinrc-xt60-battery-lead

ⓘ **NOTE** The capacitor's own leads (ø0.6 ± 0.05 mm, lead space 5.0 ± 0.5 mm). Keep them short: the capacitor lies on the air unit behind the ESC, + lead to the + pad. — panasonic-eeu-fr1v471-leads

7 Connect the air unit

links: dji_fc, dji_o4



Ⓐ ×1
DJI O4 3-in-1 cable

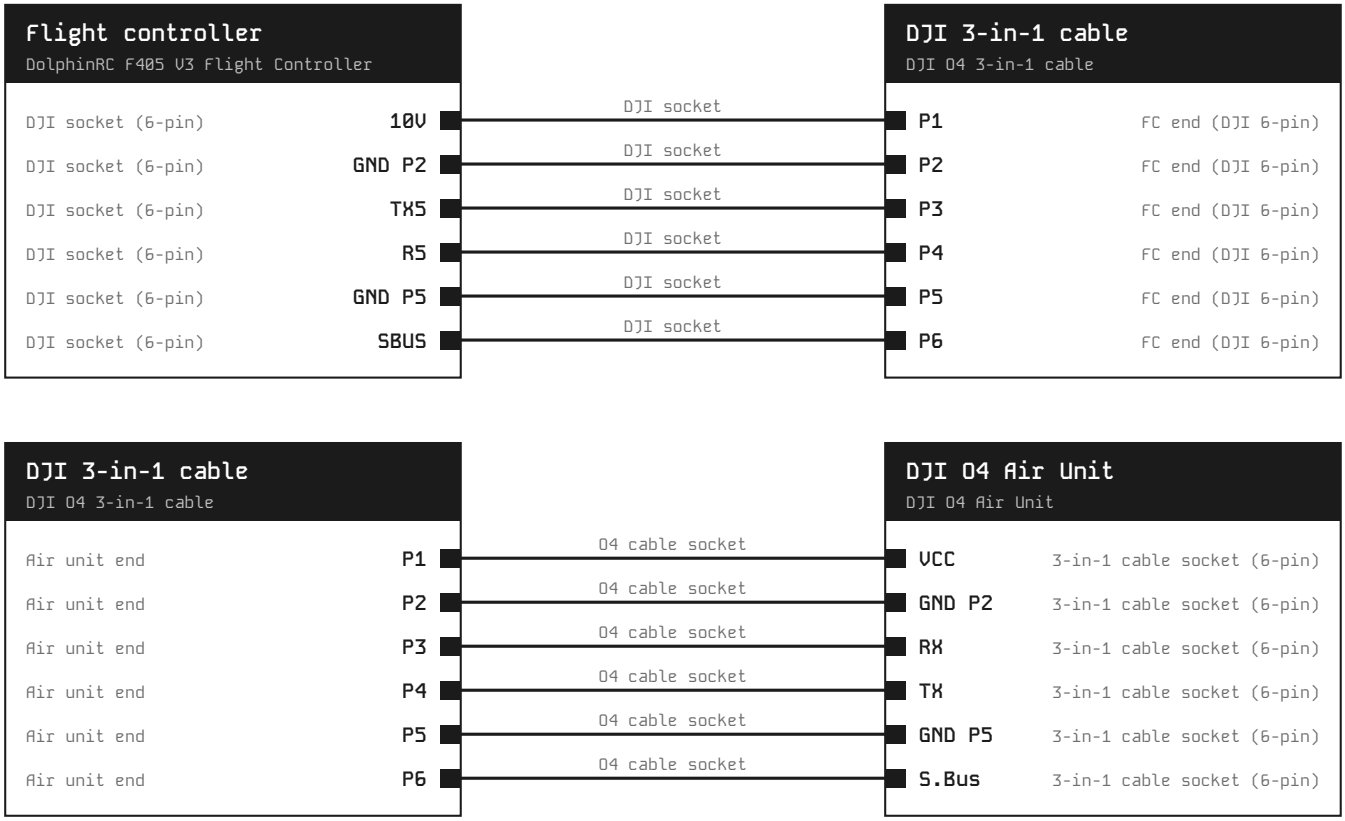


Figure 7 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Connect DJI socket, O4 cable socket as in the table (12 connections).

LINK	FROM	TO	TERMINATION	DRC
DJI socket	Flight controller **10V · GND P2 · TX5 · R5 · GND P5 · SBUS**	DJI 3-in-1 cable **P1 · P2 · P3 · P4 · P5 · P6**	dji_6pin / dji_6pin	CONFIGURED
O4 cable socket	DJI 3-in-1 cable **P1 · P2 · P3 · P4 · P5 · P6**	DJI O4 Air Unit **VCC · GND P2 · RX · TX · GND P5 · S.Bus**	dji_3in1_cable_6pin / dji_3in1_cable_6pin	CONFIGURED

Step 7 of 12

NOTE Ships with the DJI O4 Air Unit. Wire order VCC (red), GND (black), RX (white), TX (grey), GND (brown), S.Bus (yellow) (O4 user manual). DolphinRC also includes a 150 mm DJI 6-pin cable with the stack; the 50 mm DJI cable reaches here because the air unit sits right behind the stack. — dji-o4-3in1-cable

8 Connect the receiver

links: rx_pads, rx_fc_pads



(A) ×1
RadioMaster RP1 V2 ExpressLRS 2.4GHz Nano Receiver



(B) ×1
Receiver wires (RP1 V2 CRSF wire)

Figure 8 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Plug the receiver's T antenna into its U.FL socket before soldering.
2. Connect Receiver pads, Receiver wires on the FC as in the table (8 connections).

LINK	FROM	TO	TERMINATION	DRC
Receiver pads	receiver **TX · RX · 5V · G**	Receiver wires **P1 · P2 · P3 · P4**	solder_leads	CONFIGURED
Receiver wires on the FC	Receiver wires **P1 · P2 · P3 · P4**	stack **RX2 · TX2 · 5V · GND**	solder_leads	CONFIGURED

Step 8 of 12

NOTE Mounting: none (no holes); heat shrink / foam tape / zip tie. — radiomaster-rp1-v2-elrs-2g4

NOTE Package includes '1 * CRSF wire': four silicone wires, red, black, white and green, twisted, with bare tinned ends and no plug (product page and photo). Receiver TX to the FC's RX2, receiver RX to TX2 (ExpressLRS wiring guide). — radiomaster-rp1-crsf-wire

9 Fit the top plate

mates: top_plate_mount

	A ×1	5-inch frame top plate
	B ×8	M3 × 8 mm SHCS
	C ×4	M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF

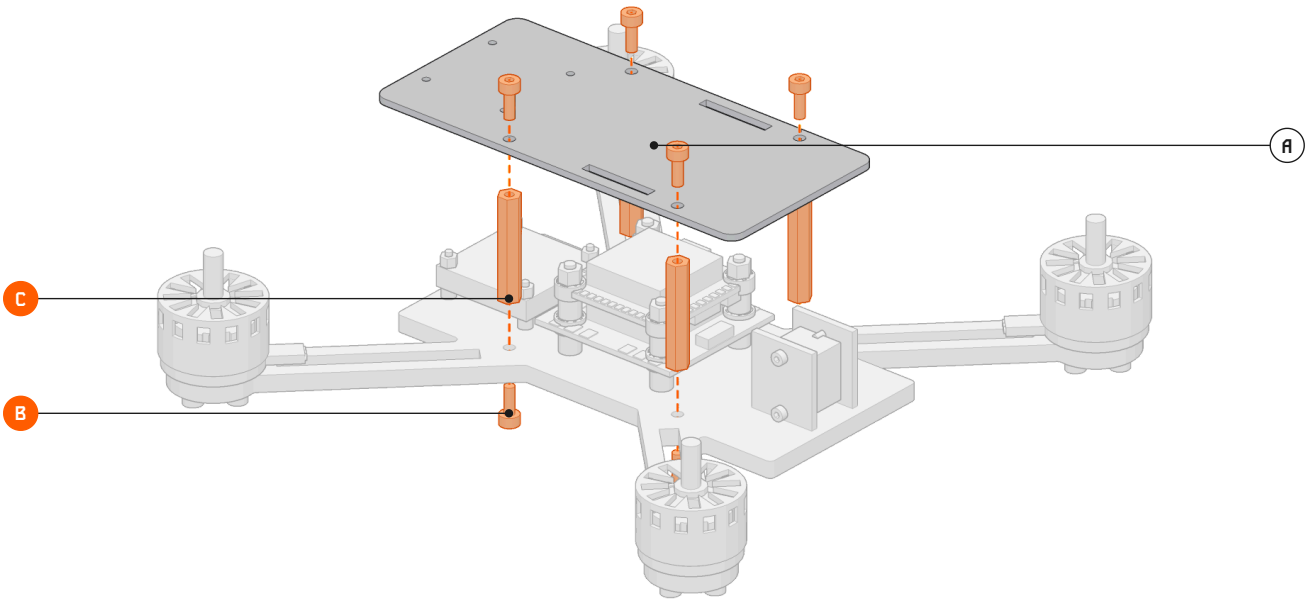


Figure 9 Fit the top plate: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Slide the clear heat-shrink sleeve over the soldered receiver and shrink it, leaving the antenna lead and the wires out of the ends.
2. Stick the sleeved receiver to the underside of the top plate at the rear right with a 10 × 10 mm piece of foam tape, so it sits there when the plate goes on (it is taped, not mounted).
3. Insert 4 × M3 × 8 mm SHCS from below, up through the top-plate standoffs.
4. Thread 4 × M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF onto the screws.
5. Lower the 5-inch frame top plate onto the top-plate standoffs (30 mm gap).
6. Drive 4 × M3 × 8 mm SHCS from above, head down.
7. Tie the receiver's T antenna to a rear standoff with a cable tie, its two elements out of the rear of the frame and clear of the propellers.

Torque 0.6 N·m[A] M3 × 8 mm SHCS **Tools** 2.5 mm hex key, 5.5 mm spanner **Step** 9 of 12

TIP Threadlocker: medium strength (blue) on the standoff screws (lower and upper). Steel into aluminium with no locking element, so vibration can back the screws out. Advice is from X-Team's motor-screw guide, a secondary source; no maker of these parts states it. — quadcopter-5in-frame-hardware

10 Mount and connect the GNSS

mates: gps_mount · links: gnss_socket, gnss_pads



A ×1
Matek M9N-5883 GNSS & Compass



B ×1
GNSS lead (GH6P, FC end soldered)



C ×4
M2 nyloc nut, DIN 985, A2 stainless



D ×4
M2 nylon spacer, Ø2.7/4.0 × 5 mm



E ×4
M2 × 12 mm SHCS

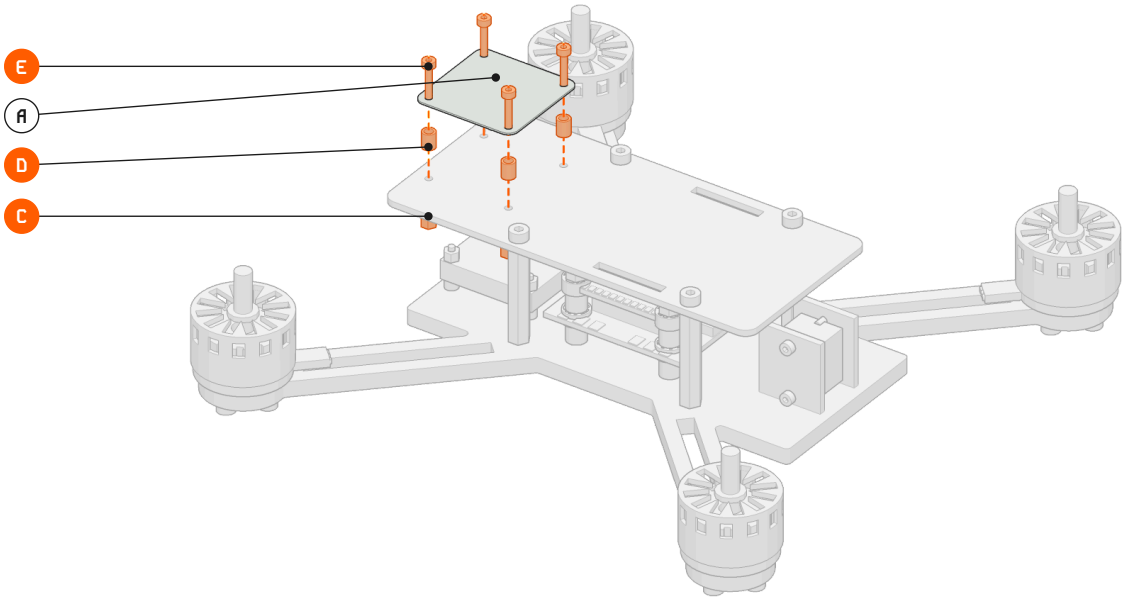


Figure 10 Mount and connect the GNSS: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Plug the GH6P lead into the GNSS's rear socket (GH6P-2) and lead its cut end down between the plates to the FC's right-edge pads.
2. Hold 4 × M2 nyloc nut, DIN 985, A2 stainless under the GPS mount.
3. Fit 4 × M2 nylon spacer, Ø2.7/4.0 × 5 mm on the GPS mount.
4. Lower the Matek M9N-5883 GNSS & Compass onto the GPS mount (5 mm gap).
5. Drive 4 × M2 × 12 mm SHCS from above, head down.
6. Connect GH6P-2, GNSS pads as in the table (12 connections).

LINK	FROM	TO	TERMINATION	DRC
GH6P-2	GNSS + compass **5V · RX · TX · CL · DA · G**	GNSS lead **P1 · P2 · P3 · P4 · P5 · P6**	jst_gh_6 / jst_gh_6	CONFIGURED
GNSS pads	GNSS lead **P1 · P2 · P3 · P4 · P5 · P6**	stack **4.5V · TX1 · RX1 · SCL · SDA · GND**	solder_leads	CONFIGURED

Torque 0.1 N·m[A] M2 × 12 mm SHCS **Tools** 1.5 mm hex key, 4 mm nut driver **Step** 10 of 12

TIP Arrow forward: the compass alignment in the configuration (align_mag) assumes the module's arrow points to the nose.

NOTE Included with the M9N-5883 ('JST-GH-6P to JST-GH-6P 20cm silicon wire'). Cut one plug off and solder to the FC pads: GNSS RX to TX1, GNSS TX to RX1. Plugged into GH6P-2, the socket facing the rear of the top plate (GH6P-1 faces the battery, 5 mm away). — matek-gh6p-gnss-lead

11 Strap on the battery



(A) ×1
CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60)

(B) ×1
HGLRC Kevlar battery strap 20x250mm

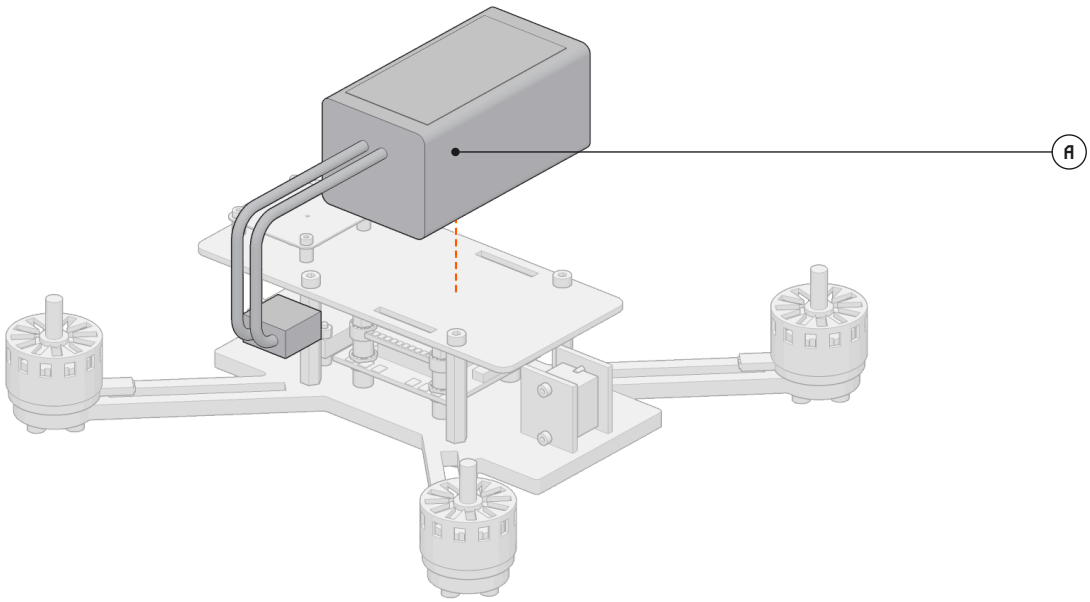


Figure 11 Strap on the battery: shown lifted above its seat.

1. Thread the strap down through one top-plate slot, under the plate, up through the other slot (the slots sit either side of the pack's centre).
2. Place the CNHL Black Series 1100mAh 6S 22.2V 100C LiPo (XT60) as shown. It has no mechanical interface in the model; the figure places it by the frame's battery pad: centred on the top plate between the strap slots, its discharge lead at the rear.
3. Fit the HGLRC Kevlar battery strap 20x250mm.
4. Close the strap over the pack through its buckle and press the hook and loop down; the pack must not slide by hand.

Step 11 of 12

⚠ **CAUTION** Fit and connect the battery only for flight; remove the props for any bench work.

ⓘ **NOTE** Mounting: none — strapped around the body (no strap interface vocabulary). — cnhl-black-series-1100mah-6s-100c

12 Fit the propellers

mates: arm_fr/prop_on_shaft, arm_rr/prop_on_shaft, arm_rl/prop_on_shaft, arm_fl/prop_on_shaft



(A) ×4
HQProp Ethix S5 5x4x3 Propeller



(B) ×4
M5 flanged lock nut (prop nut, included with the motor)

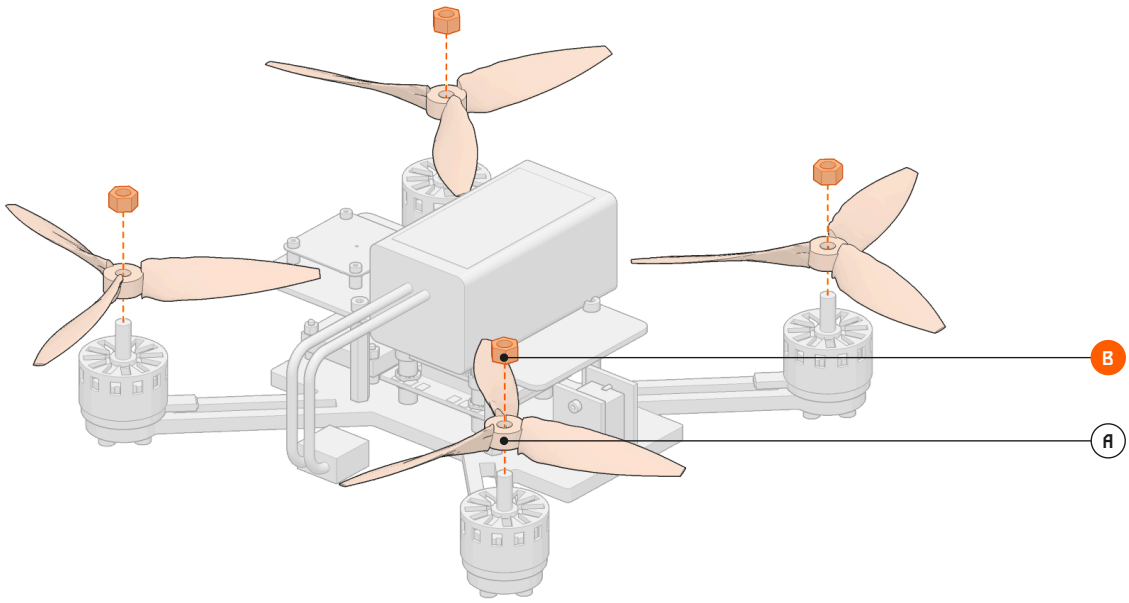


Figure 12 Fit the propellers: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Seat a HQProp Ethix S5 5x4x3 Propeller onto each output shaft Ø5 mm M5.
2. Finish with 1 × M5 flanged lock nut (prop nut, included with the motor); tighten until snug. Repeat on all 4.
3. Front-right arm: motor spins CCW from above, so fit a **CCW** HQProp Ethix S5 5x4x3 Propeller.
4. Rear-right arm: motor spins CW from above, so fit a **CW** HQProp Ethix S5 5x4x3 Propeller.
5. Rear-left arm: motor spins CCW from above, so fit a **CCW** HQProp Ethix S5 5x4x3 Propeller.
6. Front-left arm: motor spins CW from above, so fit a **CW** HQProp Ethix S5 5x4x3 Propeller.

Torque 1 N·m[A] M5 flanged lock nut (prop nut, included with the motor) **Tools** 8 mm nut driver **Step** 12 of 12

⚠ **WARNING** Fit propellers last, after configuration and a motor-direction check with the props off.

⚠ **CAUTION** A pack holds 2 CW and 2 CCW props; the steps above say which goes on each motor. The directions are Betaflight's Quad-X default, props in (firmware/betaflight/cli.txt sets yaw_motors_reversed to match). Check each motor's direction with the props off before fitting them.

ⓘ **NOTE** Tighten until the hub is clamped and cannot slip on the shaft, holding the motor bell, then stop: the polycarbonate hub cracks if crushed. No motor or propeller maker states a torque; the stack's 1.0 N·m is an assumption. Check the nut before every flight. The nylon insert locks it. — quadcopter-5in-prop-hardware

16 Coverage

ASSEMBLE() · WIRINGCHECKLIST

Every rigid mate, every hardware placement and every electrical link in the model, and the step that covers it. The verifier fails the guide if any row is not covered.

MATES

MATE	INTERFACES	HARNESS	STEP
arm_fr_mount	arm_fr/motor.base_mount ↔ frame.motor_mount_fr	arm_fr_hardware	1 · Mount the motors
arm_rr_mount	arm_rr/motor.base_mount ↔ frame.motor_mount_rr	arm_rr_hardware	1 · Mount the motors
arm_rl_mount	arm_rl/motor.base_mount ↔ frame.motor_mount_rl	arm_rl_hardware	1 · Mount the motors
arm_fl_mount	arm_fl/motor.base_mount ↔ frame.motor_mount_fl	arm_fl_hardware	1 · Mount the motors
stack_mount	stack/esc.stack_mount ↔ frame.stack_mount	stack_hardware	3 · Mount the stack on the frame
fc_mount	stack/fc.stack_mount ↔ frame.stack_mount	stack_hardware	3 · Mount the stack on the frame
top_plate_mount	frame.standoff_mount ↔ top_plate.standoff_mount	frame_hardware	9 · Fit the top plate
camera_mount	video.camera_mount ↔ frame.camera_mount	camera_hardware	5 · Install the air unit and camera
vtx_mount	video.tx_module_mount ↔ frame.vtx_mount	vtx_hardware	5 · Install the air unit and camera
gps_mount	gnss.mount ↔ top_plate.gps_mount	gps_hardware	10 · Mount and connect the GNSS
arm_fr/prop_on_shaft	arm_fr/motor.shaft ↔ arm_fr/prop.hub_bore	prop_hardware	12 · Fit the propellers
arm_rr/prop_on_shaft	arm_rr/motor.shaft ↔ arm_rr/prop.hub_bore	prop_hardware	12 · Fit the propellers
arm_rl/prop_on_shaft	arm_rl/motor.shaft ↔ arm_rl/prop.hub_bore	prop_hardware	12 · Fit the propellers
arm_fl/prop_on_shaft	arm_fl/motor.shaft ↔ arm_fl/prop.hub_bore	prop_hardware	12 · Fit the propellers

HARDWARE PLACEMENTS

HARNESS / CHILD	PARTS	JOINT	STEP
arm_fr_hardware/screws	4 × M3 x 8 mm motor screw	arm_fr_mount	1 · Mount the motors
arm_rr_hardware/screws	4 × M3 x 8 mm motor screw	arm_rr_mount	1 · Mount the motors
arm_rl_hardware/screws	4 × M3 x 8 mm motor screw	arm_rl_mount	1 · Mount the motors
arm_fl_hardware/screws	4 × M3 x 8 mm motor screw	arm_fl_mount	1 · Mount the motors
stack_hardware/screws	4 × M3 x 30 mm SHCS	stack_mount	3 · Mount the stack on the frame
stack_hardware/spacers	8 × ETTINGER 005.83.060 M3 round spacer 6 mm, PA6 GF25	stack_mount	3 · Mount the stack on the frame
stack_hardware/esc_grommets	4 × M3 rubber grommet, ESC	stack_mount	3 · Mount the stack on the frame
stack_hardware/fc_grommets	4 × M3 rubber grommet, FC	stack_mount	3 · Mount the stack on the frame
stack_hardware/nuts	4 × M3 nyloc nut, ISO 10511 / DIN 985 Type T, A2 stainless	stack_mount	3 · Mount the stack on the frame
frame_hardware/screws	8 × M3 x 8 mm SHCS	top_plate_mount	9 · Fit the top plate
frame_hardware/standoffs	4 × M3 aluminium standoff, 30 mm, female–female, 5.5 mm AF	top_plate_mount	9 · Fit the top plate
camera_hardware/screws	4 × M2 x 5 mm SHCS	camera_mount	5 · Install the air unit and camera
vtx_hardware/screws	4 × M2 x 18 mm SHCS	vtx_mount	5 · Install the air unit and camera

HARDWARE (CONTINUED)

HARNESS / CHILD	PARTS	JOINT	STEP
vtx_hardware/spacers	4 × M2 nylon spacer, Ø2.7/4.0 × 3 mm	vtx_mount	5 · Install the air unit and camera
vtx_hardware/nuts	4 × M2 nyloc nut, DIN 985, A2 stainless	vtx_mount	5 · Install the air unit and camera
gps_hardware/nuts	4 × M2 nyloc nut, DIN 985, A2 stainless	gps_mount	10 · Mount and connect the GNSS
gps_hardware/spacers	4 × M2 nylon spacer, Ø2.7/4.0 × 5 mm	gps_mount	10 · Mount and connect the GNSS
gps_hardware/screws	4 × M2 × 12 mm SHCS	gps_mount	10 · Mount and connect the GNSS
arm_fr/prop_hardware/nut	1 × M5 flanged lock nut (prop nut, included with the motor)	arm_fr/prop_on_shaft	12 · Fit the propellers
arm_rr/prop_hardware/nut	1 × M5 flanged lock nut (prop nut, included with the motor)	arm_rr/prop_on_shaft	12 · Fit the propellers
arm_rl/prop_hardware/nut	1 × M5 flanged lock nut (prop nut, included with the motor)	arm_rl/prop_on_shaft	12 · Fit the propellers
arm_fl/prop_hardware/nut	1 × M5 flanged lock nut (prop nut, included with the motor)	arm_fl/prop_on_shaft	12 · Fit the propellers

WIRING LINKS

LINK	NAME	HARNESS	STEP
battery_plug	XT60	—	6 · Solder the battery lead and capacitor
battery_pads	ESC battery pads	—	6 · Solder the battery lead and capacitor
bulk_cap_can	Capacitor leads	—	6 · Solder the battery lead and capacitor
bulk_cap_pads	Capacitor on the ESC battery pads	—	6 · Solder the battery lead and capacitor
m2_leads	Motor 2 leads	—	4 · Solder the motor leads
m2_pads	Motor 2 pads	—	4 · Solder the motor leads
m1_leads	Motor 1 leads	—	4 · Solder the motor leads
m1_pads	Motor 1 pads	—	4 · Solder the motor leads
m3_leads	Motor 3 leads	—	4 · Solder the motor leads
m3_pads	Motor 3 pads	—	4 · Solder the motor leads
m4_leads	Motor 4 leads	—	4 · Solder the motor leads
m4_pads	Motor 4 pads	—	4 · Solder the motor leads
rx_pads	Receiver pads	—	8 · Connect the receiver
rx_fc_pads	Receiver wires on the FC	—	8 · Connect the receiver
dji_fc	DJI socket	—	7 · Connect the air unit
dji_o4	O4 cable socket	—	7 · Connect the air unit
gnss_socket	GH6P-2	—	10 · Mount and connect the GNSS
gnss_pads	GNSS pads	—	10 · Mount and connect the GNSS
stack/fc_esc_plug_fc	FC ESC socket	—	2 · Connect the FC and ESC
stack/fc_esc_plug_esc	ESC FC socket	—	2 · Connect the FC and ESC